Pushkar Verma

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EDUCATION

International Institute of Information Technology, Bangalore (2023 – 2024)
Executive Post Graduate Programme in **Machine Learning and Artificial Intelligence**

Bangalore Institute of Technology, Bangalore (2015 – 2019)
Bachelor of Engineering in **Electronics & Communication Engineering**

PROFESSIONAL EXPERIENCE

Medi Assist Healthcare Services Ltd. | *Lead Machine Learning Engineer* | Feb 2025 – Present

- **Intelligent Document Processing (IDP):** Led the end-to-end development of a multi-stage IDP pipeline (orientation correction → OCR-based classification → Vision-Language Model structured extraction) currently live for Pre-Authorizations and in shadow testing for cashless claims.
- **Enterprise Agentic AI Assistant:** Architected a company-wide LangGraph AI assistant with strict privacy guardrails and row-based access controls (RBAC), featuring an NL-to-SQL enhancer agent for custom reporting and RAG-based KPI/dashboard routing to automate ~85% of report requests by volume.
- **Out-of-Pocket (OOP) & Risk Prediction:** Designed a novel 4-model architecture to predict OOP expenses and classify high-risk cases for early patient discharges, boosting leads by 20% and reducing MAE by 40%.
- **Bill Header Decomposer:** Engineered a historical bill-trained decomposition model to break claimed amounts into insurance-specific headers, directly feeding the Rule Engine to automate Pre-Auth approvals and downstream workflows.
- **ICD & Treatment Classification:** Built an ensemble classification engine (Regex fast-path, fine-tuned LLM, and hospital-specific similarity search) to map extracted Claim Form and Discharge Summary fields to ICD codes and validate treatment-ailment compatibility.
- **Customer Support Chatbot:** Architected a deterministic Dialogflow CX and Python chatbot leveraging Rule Engine tool-calling for policy FAQ/coverage queries and demographic-based (age/gender) ailment and treatment recommendations; currently live for support teams prior to end-user rollout.

Ericsson India Private Ltd. | *Data Scientist* | Sep 2019 – Feb 2025

- **RAN Thermal Forecasting:** Developed a 3-day max cabinet temperature forecasting model using site configuration, weather, and telemetry data to enable proactive hardware problem management.
- **Incident Ticket Classification:** Built an NLP classification model using ticket topics and descriptions, achieving 90% accuracy and significantly reducing resolution turnaround times.
- **Real-Time Telecom Analytics:** Built live KPI and User Experience monitoring dashboards to support Ooredoo Qatar SOC teams during the FIFA World Cup 2022, and delivered cross-functional ML solutions across RAN, Transmission, and Incident/Problem Management.
- **Technical Mentorship:** Conducted 17 theoretical and practical ML training sessions within the Business Unit to prepare engineers for Intermediate and Advanced ML Certification programs.

TECHNICAL SKILLS

Generative AI & LLMs: Agentic AI, RAG, LangGraph, LangChain, Google ADK, LoRA Fine-Tuning, Ollama (Self-Hosted LLMs), Vision-Language Models, LayoutLM.

ML & Deep Learning: Scikit-learn, XGBoost, CatBoost, LightGBM, PyTorch, TensorFlow, Keras, Hugging Face, Pandas, NumPy, Matplotlib, Seaborn, MLflow.

ML Domains: NLP, Regression, Classification, Transformers & Attention, Autoencoders, Anomaly Detection, Explainable AI (SHAP, Permutation Importance), Recommendation Systems (Matrix Factorization).

Cloud & MLOps (GCP): Vertex AI, AI Platform, BigQuery, Bigtable, Cloud Run, Cloud Build, Model Registry, RAG Engine, Pub/Sub, Firestore, Docker, FastAPI, Git.

PERSONAL PROJECTS

Quantamental_SwingTrading_Strategy | Live – Stock picks every quarter using Fundamental & Technical Indicators.

Stumped-AI | Live – An IPL Stats agent that created NL-SQL Queries and creates plots in Python.